

MICROCANONICAL PREPARATION AND EXPONENTIAL-SCALE ENSEMBLE TRANSFER FOR DETERMINISTIC HARD SPHERES

QIAN QI

ABSTRACT. We prove source-dependent microcanonical preparation at the exact finite-volume saddle. The grand-canonical logarithm is decomposed into its true compound-Poisson singleton exponent and a connected remainder; smooth convolution powers arise only after exponentiating the singleton term. Under explicit lattice-span, rank, and nonlattice hypotheses this yields a uniform mixed characteristic-function theorem and exact-number shrinking-shell local limit. The conditioned pressure, initial rate, and fluctuation covariance are the source-dependent constrained infimum and its Schur complement.

Platform identifier: HSBG-LI0-v2. **Controlling revision:** ROUND6-REFEREE-POSITIVE-CLOSURE.

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1. ROUND-SIX POSITIVE CLOSURE: COMPOUND-POISSON SMOOTHING AT THE EXACT FINITE SADDLE

The conditioning change of measure is centered at the exact finite-volume mean. Smoothing of momentum and energy is derived from the singleton compound-Poisson exponent of the grand-canonical logarithm; no collection of independent particles is inserted as an additional connected cluster.

Date: August 31, 2026.

2020 *Mathematics Subject Classification.* Primary 82C40, 60F10; Secondary 76P05, 82C22.

Key words and phrases. hard spheres, microcanonical ensemble, Liouville measure, information projection, Boltzmann-Grad limit.

1.1. Constraint coordinates and the finite pressure. Let

$$C(x, v) = \left(1, v, \frac{|v|^2}{2}, \chi_1(x, v), \dots, \chi_m(x, v) \right) \in \mathbb{R}^d.$$

The first coordinate is lattice valued after multiplication by the large-deviation speed μ_ε ; the remaining coordinates are continuous. We impose the following explicit nondegeneracy conditions.

- (C1) the particle-number lattice has span one;
- (C2) after conditioning on number, the pushforward of the singleton activity by the continuous part of C has a C^{d+4} density on one open set and its affine span is $\mathbb{R}^{d-1}$;
- (C3) no nonzero continuous frequency has modulus-one singleton characteristic function on all positive-activity patches;
- (C4) the target vector lies in a compact subset of the interior mean image.

For a particle-path/contact source H and multiplier λ , put

$$Q_\varepsilon(H, \lambda) = \frac{1}{\mu_\varepsilon} \log \mathbb{E}_\varepsilon^{\text{gc}} \exp \left\{ \mu_\varepsilon \langle H, (\Pi^\varepsilon, \Gamma^\varepsilon) \rangle + \lambda \cdot \sum_i C(z_i(0)) \right\}.$$

The B2 grand-canonical theorem supplies a normal connected expansion on every bounded real source ball and on a complex neighborhood of its interior.

1.2. The exact singleton exponent. Let $a_{H,\lambda}^\varepsilon(dz)$ be the one-label activity after the one-particle part of the dynamic source has been integrated along a free trajectory. The connected logarithm separates exactly as

$$Q_\varepsilon(H, \lambda + i\theta) - Q_\varepsilon(H, \lambda) = \mathfrak{p}_{1,\varepsilon}(\theta) + \mathfrak{r}_\varepsilon(H, \lambda, \theta),$$

where

$$\mathfrak{p}_{1,\varepsilon}(\theta) = \int \left(e^{i\theta \cdot C(z)} - 1 \right) a_{H,\lambda}^\varepsilon(dz)$$

is the compound-Poisson singleton exponent and $\mathfrak{r}_\varepsilon$ is the sum of connected clusters with at least two labels.

Lemma 1.1 (Compound-Poisson smoothing and connected remainder). *On compact source and multiplier sets there are constants $c, C, \delta > 0$ such that:*

- (i) *the singleton activity has total mass bounded above and below and has uniform C^{d+4} densities on the patches in (C2);*
- (ii) *for $|\theta| \leq \delta$,*

$$\Re \mathfrak{p}_{1,\varepsilon}(\theta) \leq -c|\theta|^2;$$

- (iii) *on every compact frequency set separated from the exact mixed lattice origin,*

$$\Re \mathfrak{p}_{1,\varepsilon}(\theta) \leq -c;$$

- (iv) *for continuous frequencies $|u| \geq 1$, the exact compound-Poisson coefficient obtained from*

$$\exp\{\mu_\varepsilon \mathfrak{p}_{1,\varepsilon}(t, u)\} = e^{-\mu_\varepsilon a_1} \sum_{n \geq 0} \frac{\mu_\varepsilon^n}{n!} \left(\hat{a}_{H,\lambda}^\varepsilon(t, u) \right)^n$$

has an integrable Fourier majorant of arbitrary prescribed polynomial order;

(v) the connected remainder and its first four frequency derivatives obey

$$|\partial_\theta^j \mathbf{r}_\varepsilon| \leq C(T + \varepsilon), \quad 0 \leq j \leq 4.$$

Proof. The one-label term in a grand-canonical logarithm is exactly the singleton activity integral. Exponentiating it gives the displayed Poisson series; this is where convolution powers of singleton marks occur. They are not added to the connected logarithm. Conditions (C2)–(C3) and compactness give the quadratic and compact-frequency Cramer bounds for the singleton characteristic function. A convolution power of order at least $d + 4$ has an integrable C^4 density by the coarea rank condition. Repeated integration by parts therefore gives arbitrary polynomial Fourier decay for the corresponding terms of the exact Poisson series. The finitely many smaller powers are handled directly and the Poisson tail is exponentially summable.

Every cluster in $\mathbf{r}_\varepsilon$ contains at least one connecting interaction or contact and hence one factor $T + \varepsilon$ in the B2 trace-class cluster norm. Mark insertions differentiate only bounded constraint coordinates on the declared source chart. Normal convergence gives the uniform derivative bound. Decreasing the common short-time horizon, the singleton gaps dominate the connected remainder. $\square$

1.3. Uniform finite-volume convexity and mean image.

Lemma 1.2 (Number/mark covariance block). *The finite-volume Hessian admits the block decomposition*

$$D_{\lambda\lambda}^2 Q_\varepsilon = \begin{pmatrix} \sigma_N^2 & b^T \\ b & \Sigma_Y \end{pmatrix},$$

where $\sigma_N^2 \geq c$ comes from activity fluctuations and the Schur complement $\Sigma_Y - bb^T/\sigma_N^2 \geq cI$. Hence

$$cI \leq D_{\lambda\lambda}^2 Q_\varepsilon(H, \lambda) \leq cI$$

uniformly on compact source charts.

Proof. The constant coordinate is not centered inside a fixed-particle singleton law; its fluctuation is the compound-Poisson activity count. This gives the positive number block. Conditional on the number sector, the continuous mark covariance is positive by (C2) and the spanning activity patches. The Schur complement is therefore positive for the singleton exponent. The connected correction to the Hessian is $O(T + \varepsilon)$ by Lemma 1.1; choose the common horizon so it is smaller than half the singleton lower bound. Upper bounds follow from Gaussian velocity tails and normal convergence. $\square$

Let a_ε be the exact finite shell center, including the integer particle number divided by μ_ε .

Theorem 1.3 (Exact finite-volume saddle). *For every source H in a compact source ball and every compatible target a_ε in the fixed interior set, there is a unique multiplier $\lambda_{H,\varepsilon}$ such that*

$$D_\lambda Q_\varepsilon(H, \lambda_{H,\varepsilon}) = a_\varepsilon.$$

The multipliers remain in one compact interior chart, converge locally uniformly to the unique limiting saddle λ_H , and are analytic in H .

Proof. The limiting gradient maps the compact multiplier chart diffeomorphically onto an open set containing the target compact set. Uniform convergence of the finite gradients on the chart places the same compact target inside every finite mean image for small ε . The uniform Hessian lower bound from Lemma 1.2 gives injectivity and the quantitative inverse-function theorem gives existence, uniqueness, and analytic source dependence. Compactness and uniqueness identify the limit. $\square$

1.4. Mixed lattice–continuous local limit at the finite saddle. Under the $(H, \lambda_{H,\varepsilon})$ -tilted law, write

$$\hat{N} = N - N_\varepsilon, \quad \hat{Y} = Y - \mathbb{E}_{H, \lambda_{H,\varepsilon}} Y.$$

By construction both means are exactly centered at the target shell.

Theorem 1.4 (Uniform mixed characteristic-function theorem). *There are $c, \delta > 0$ such that the normalized joint characteristic function $\varphi_\varepsilon(t, u)$ satisfies*

$$|\varphi_\varepsilon(t, u)| \leq \begin{cases} \exp\{-c\mu_\varepsilon(t^2 + |u|^2)\}, & |t| + |u| \leq \delta, \\ \exp\{-c\mu_\varepsilon\}, & \delta \leq d((t, u), \Lambda^*) \leq C, \\ C_M(1 + \sqrt{\mu_\varepsilon}|u|)^{-M}, & |u| \geq C, \end{cases}$$

where $t \in [-\pi, \pi]$, Λ^ is the exact mixed dual lattice, and the fundamental domain contains only the origin. The estimates and their first two source derivatives are uniform on compact source and target sets.*

Proof. Near zero, Taylor expansion at the exact finite saddle and Lemma 1.2 give the Gaussian bound with no mean error. On compact minor arcs, Lemma 1.1 gives a strict singleton gap. The connected remainder is smaller on the common short horizon. The span-one and nonlattice assumptions identify the origin as the only dual lattice point. For large continuous frequency, use the exact Poisson series and its smooth convolution powers from Lemma 1.1; the connected factor has bounded derivatives and is absorbed by convolution. Source derivatives insert only integrable marks, so the same majorants apply. $\square$

Let $B \subset \mathbb{R}^{d-1}$ be a fixed bounded convex set with piecewise C^2 boundary and nonempty interior. Let $\delta_\varepsilon \downarrow 0$ satisfy

$$\sqrt{\mu_\varepsilon} \delta_\varepsilon \rightarrow \infty.$$

Theorem 1.5 (Exact-number shrinking-shell coefficient). *Uniformly on compact source and target sets,*

$$\mathbb{P}_{H,\lambda_H,\varepsilon} \{N = N_\varepsilon, Y - a'_\varepsilon \in \delta_\varepsilon B\} = \mu_\varepsilon^{-1/2} (c_{H,a} + o(1)),$$

where $c_{H,a}$ is continuous and bounded above and below by positive constants. More generally, if the continuous shell is on the Gaussian scale, its limiting Gaussian mass replaces one; for narrower shells the corresponding mixed local-density formula holds under the usual boundary-smoothing scale. In every case used for microcanonical conditioning, the shell logarithm is $o(\mu_\varepsilon)$.

Proof. Use Fourier inversion in the exact number coordinate and smooth upper and lower approximations to $1_{\delta_\varepsilon B}$. In the central region, expand the pressure at the exact finite saddle. The lattice coefficient gives the Schur-complement one-dimensional factor $(2\pi\mu_\varepsilon\sigma_{N|Y}^2)^{-1/2}$. The continuous shell contains an increasing number of standard deviations and its Gaussian mass tends to one. Boundary smoothing contributes $o(\mu_\varepsilon^{-1/2})$ because the boundary is piecewise C^2 and the smoothing width can be chosen between $\mu_\varepsilon^{-1/2}$ and δ_ε . The three remaining Fourier regions are negligible by Theorem 1.4. Uniform covariance bounds give the positive bounds on $c_{H,a}$. $\square$

1.5. Source-dependent conditioning. Let $\mathcal{S}_\varepsilon(a_\varepsilon)$ denote the exact number and continuous shell above. Define

$$Q_\varepsilon^{\text{mc},a}(H) = \frac{1}{\mu_\varepsilon} \log \mathbb{E}_\varepsilon^{\text{gc}} \left[e^{\mu_\varepsilon \langle H, X_\varepsilon \rangle} \mid \mathcal{S}_\varepsilon(a_\varepsilon) \right].$$

Theorem 1.6 (Microcanonical pressure and covariance). *Locally uniformly in H ,*

$$Q_\varepsilon^{\text{mc},a}(H) \longrightarrow Q(H, \lambda_H) - \lambda_H \cdot a - Q(0, \lambda_0) + \lambda_0 \cdot a.$$

Equivalently this is the difference of the two constrained infima in λ . Its Hessian is

$$Q_{HH} - Q_{H\lambda} Q_{\lambda\lambda}^{-1} Q_{\lambda H},$$

the exact constraint Schur complement. For a time-zero constraint source H_η , the pressure equals $\eta \cdot a$.

Proof. Change measure in the numerator by the exact finite-volume tilt $(H, \lambda_{H,\varepsilon})$. On the shell, the multiplier term is $-\mu_\varepsilon \lambda_{H,\varepsilon} \cdot a_\varepsilon + o(\mu_\varepsilon)$, and Theorem 1.5 makes the remaining shell probability subexponential. Repeat with $H = 0$ for the denominator and pass to the uniform saddle limit. Differentiating the finite saddle equation and then the envelope gives the Schur complement. For a time-zero source, $Q(H_\eta, \lambda) = Q(0, \lambda + \eta)$, so the infimum translates exactly. $\square$

1.6. The prepared initial LDP. Use the separating class of bounded time-zero cylinder sources together with a small quadratic velocity chart. Maxwellian exponential tightness gives compact empirical-measure sublevels.

Theorem 1.7 (Round-six B1 closure). *The primitive hard-sphere shell is conditioned at a unique exact finite-volume source-dependent saddle. The singleton compound-Poisson exponent supplies a uniform mixed lattice/nonlattice local limit, the shell coefficient has only subexponential cost, and the limiting pressure, initial constrained rate, and Gaussian covariance are the source-dependent infimum and Schur-complement formulas above.*

Proof. The mixed shell theorem and pressure transfer are Theorems 1.5 and 1.6. Exponential tightness and the conditional Laplace principle on the separating time-zero source class give the good initial LDP; its rate is the grand-canonical initial rate restricted to the constraint surface and normalized by its constrained minimum. The covariance statement was proved in Theorem 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: `qiqian@pku.edu.cn`